# Stat 145 (Multivariate Statistics)

## Lesson 3.3 Factor Analysis

## Introduction

In experimental design, the term *factor* refers to a categorical explanatory variable. In factor analysis, a factor is a **latent variable**- a variable which cannot be measured directly, such as intelligence or happiness.

Factor analysis may be said to originate with a 92-page article by Charles Spearman in the 1901 American Journal of Psychology, entitled "General intelligence, objectively determined and measured." If you believe that some people are generally smarter than others, the basic idea is quite natural. True intelligence cannot be directly observed, so it's a latent variable. However, we can observe performance on various tests and puzzles. Spearman proposed that the correlations among observable variables arise from their connection to a common factor — general intelligence.

Factor analysis may be divided into two types, commonly called **exploratory factor analysis** and **confirmatory factor analysis**. The goal is to describe and summarize the data by explaining a large number of observed variables in terms of a smaller number of latent variables (factors). The factors are the reason the observable variables have the correlations they do. Figure 1 shows the path diagram of a model with two factors and eight observable variables. A common rule is at least three observable variables for each factor. In general, the more variables for each factor, the better.

The general factor analysis model may be written as follows. Independently for $i = 1, 2, \cdots, n$, let

$$\mathbf{d}_i = \mathbf{\Lambda}\mathbf{F}_i + \mathbf{e}_i$$

where $\mathbf{d}_i$ is a $k \times 1$ observable random vector, $\mathbf{\Lambda}$ is a $k \times p$ matrix of constants, and $\mathbf{F}_i$ (F for factor) is a $p \times 1$ latent random vector with covariance matrix $\mathbf{\Phi}$. The $k \times 1$ vector of error terms $\mathbf{e}_i$ is independent of $\mathbf{F}_i$; it has expected value zero and covariance matrix $\mathbf{\Omega}$ which is almost always assumed to be diagonal. There are no intercepts, and $E(\mathbf{F}_i) = \mathbf{0}$. A multivariate normal assumption for $\mathbf{F}_i$ and $\mathbf{e}_i$ is common.

To clarify the notation, the model equations for Figure 1 are

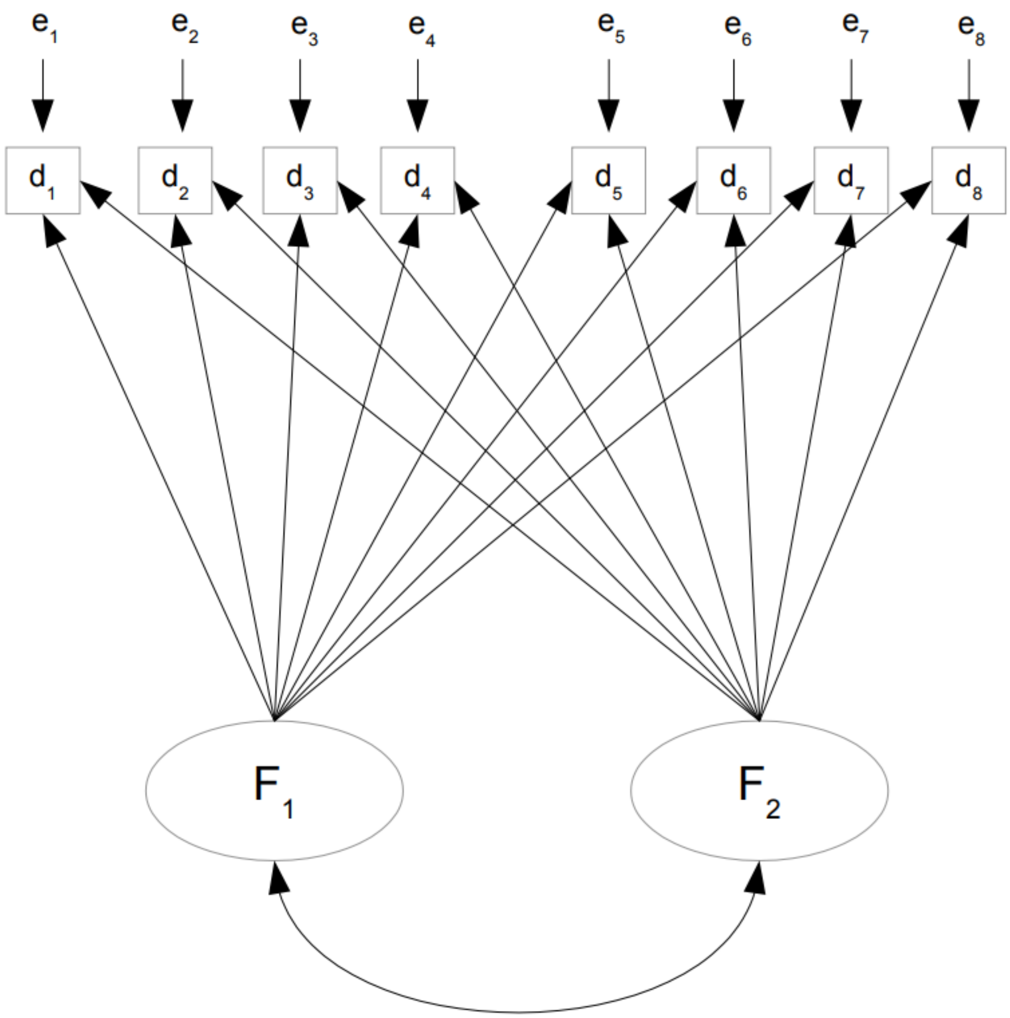



Figure 1: A Two-Factor Model

$$
\begin{array}{ccccccccc}
\mathbf{d}_i & = & \boldsymbol{\Lambda} & & \mathbf{F}_i & + & \mathbf{e}_i \\
\begin{pmatrix} d_{i,1} \\ d_{i,2} \\ d_{i,3} \\ d_{i,4} \\ d_{i,5} \\ d_{i,6} \\ d_{i,7} \\ d_{i,8} \end{pmatrix} & = & \begin{pmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \\ \lambda_{31} & \lambda_{32} \\ \lambda_{41} & \lambda_{42} \\ \lambda_{51} & \lambda_{52} \\ \lambda_{61} & \lambda_{62} \\ \lambda_{71} & \lambda_{27} \\ \lambda_{81} & \lambda_{82} \end{pmatrix} & & \begin{pmatrix} F_{i,1} \\ F_{i,2} \end{pmatrix} & + & \begin{pmatrix} e_{i,1} \\ e_{i,2} \\ e_{i,3} \\ e_{i,4} \\ e_{i,5} \\ e_{i,6} \\ e_{i,7} \\ e_{i,8} \end{pmatrix} .
\end{array}
$$

The $\lambda_{ij}$ values are called **factor loadings**. They are essentially regression coefficients linking the factors to the observed variables. The factors $F_{i,1}$ and $F_{i,2}$ are sometimes called **common factors**, because they influence all the observed variables; all the observed variables have them in common. The error terms $e_{i,1}, \cdots, e_{i,8}$ are sometimes called **unique factors**, because each one influences only a single observed variable.

The defining feature of exploratory factor analysis is that it tries to be as unconstrained as possible. The method really wants the data to speak. In Figure 1 and in general, there are arrows from all factors to all observed variables.

## Identifiability

The parameters of the general factor analysis model are massively nonidentified. That is, the parameters are cannot be uniquely estimated from a given set of data. The classical way out of this dilemma is to regard the covariance matrix of the factors as essentially arbitrary, and fix $\boldsymbol{\Phi} = \mathbf{I}$. The factors are said to be "orthogonal" (at right angles, uncorrelated). They are also standardized, meaning that the (scalar) expected value of each factor is zero, and its variance equals one. This makes interpretations simple and easy.

Setting $\boldsymbol{\Phi} = \mathbf{I}$ standardizes the factors as well as making them uncorrelated. The observed variables are standardized as well. For $j = 1, 2, \cdots, k$ and (almost) independently for $i = 1, 2, \cdots, n$ the data we work with are $z_{ij} = \frac{d_{ij} - \bar{d}_j}{s_j}$. Thus, each observed variable has variance one as well as mean zero.

In the revised exploratory factor analysis model below, the subscripts $i$ on $\mathbf{z}_i$, $\mathbf{F}_i$, and $\mathbf{e}_i$ have been dropped to reduce notational clutter. Implicitly, everything applies independently for $i = 1, 2, \cdots, n$. The model is

$$\mathbf{z} = \boldsymbol{\Lambda}\mathbf{F} + \mathbf{e}$$

where:

- $\mathbf{z}$ is a $k \times 1$ observable random vector. Each element of $\mathbf{z}$ has expected value zero and variance one.

- $\mathbf{\Lambda}$ is a $k \times p$ matrix of constants.
- $\mathbf{F}$ (F for factor) is a $p \times 1$ latent random vector with expected value zero and covariance matrix $\mathbf{I}_p$.
- The $k \times 1$ vector of error terms $\mathbf{e}$ has expected value zero and covariance matrix $\mathbf{\Omega}$, which is diagonal.

For this model, everything emerges in terms in terms of correlations rather than covariances. This is a virtue, because correlations are easier to interpret. First, $cov(\mathbf{z}_i) = \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^T + \mathbf{\Omega}$ is a correlation matrix; correspondingly, estimation and inference will be based on the sample correlation matrix.

## Factor Loadings

Next, consider the matrix of correlations between the factors and the observed variables. Because all the variables are standardized,

$$\begin{aligned}
corr(\mathbf{z}, \mathbf{F}) &= cov(\mathbf{z}, \mathbf{F}) \\
&= cov(\mathbf{\Lambda}\mathbf{F} + \mathbf{e}, \mathbf{F}) \\
&= \mathbf{\Lambda} cov(\mathbf{F}, \mathbf{F}) + cov(\mathbf{e}, \mathbf{F}) \\
&= \mathbf{\Lambda} cov(\mathbf{F}) + \mathbf{0} \\
&= \mathbf{\Lambda}\mathbf{I} \\
&= \mathbf{\Lambda}
\end{aligned}$$

Thus, the factor loadings are correlations between the observable variables and the factors. In particular, the correlation between observed variable $i$ and factor $j$ is $\lambda_{ij}$. The square of $\lambda_{ij}$ is the reliability of observed variable $i$ as a measure of factor $j$.

## Communality and Uniqueness

Observed variable $i$ (an element of $\mathbf{z}$; the index $i$ goes from $1, 2, \cdots, k$) may be written in scalar form as

$$\begin{aligned}
z_i &= \lambda_{i1} F_1 + \lambda_{i2} F_2 + \cdots + \lambda_{ip} F_p + e_i \\
&= \sum_{j=1}^{p} \lambda_{ij} F_j + e_i
\end{aligned}$$

so that

$$
\begin{aligned}
Var(z_i) &= Var\left(\sum_{j=1}^{p} \lambda_{ij} F_j + e_i\right) \\
&= \sum_{j=1}^{p} \lambda_{ij}^2 Var(F_j) + Var(e_i) \\
&= \sum_{j=1}^{p} \lambda_{ij}^2 + \omega_i
\end{aligned}
$$

where $\omega_i = Var(e_i)$ is the $i^{th}$ diagonal element of $\mathbf{\Omega}$. Since the observed variables are standardized, we have $1 = \sum_{j=1}^{p} \lambda_{ij}^2 + \omega_i$.

The variance of the observed variable has been split into two components. $\sum_{j=1}^{p} \lambda_{ij}^2$ is the proportion of variance in observed variable $i$ that comes from the common factors. It is called the **communality**. To get the communality of a variable, add up the squares of the factor loadings in the corresponding row of $\mathbf{\Lambda}$. The other component is $\omega_i = 1 - \sum_{j=1}^{p} \lambda_{ij}^2$. It is is what's left over, the part that comes from error. It is called the **uniqueness** of the variable.

In the standardized factor analysis model, the only unknown parameters are the factor loadings. This really is quite nice. Since factor loadings are the correlations between the observable variables and the factors, they could be very informative about the processes driving the data. Squared factor loadings are reliabilities, another important feature of the measurement model. One could also use estimated factor loadings to estimate how much of the variance in each observable variable comes from each factor.

## Factor Extraction

The factor extraction phase in Exploratory Factor Analysis (EFA) is the process of estimating the factor loadings and the communalities to determine the initial, unrotated factor solution.

This is the step where the statistical algorithm determines the minimum number of factors required to explain the observed correlations among the variables. The core task is to model the relationships between your measured variables using a smaller set of underlying latent variables (factors).

## Factor Rotation

Factor analysis initially extracts factors that maximize the variance explained, often resulting in complex patterns where variables load significantly on multiple factors. Rotation mathematically reorients the factor axes in the factor space, like turning a coordinate system, so that variables cluster more closely to the new axes. The desired outcome is a factor loading matrix where:

- Each variable loads highly on only one factor.
- Each factor has high loadings for only a subset of variables.

This simplification makes it clear which measured variables belong to which latent factor, facilitating the naming and interpretation of the factors.

### Types of Factor Rotation

Factor rotation methods fall into two main categories, distinguished by whether they assume the underlying factors are correlated or uncorrelated.

1. **Orthogonal Rotation** (Uncorrelated Factors): The factor axes are kept perpendicular (at a 90-degree angle) to one another. This mathematically enforces the assumption that the extracted factors are uncorrelated (independent). Thisis appropriate to use when there is a strong theoretical reason to believe the latent factors should be independent of each other (e.g., verbal and spatial reasoning are often treated as orthogonal constructs). Common methods of orthogonal rotation are:

- **Varimax**: The most popular method. It simplifies the columns of the factor loading matrix, maximizing the variance of squared loadings for each factor. This tends to make each factor have a few high loadings and many near-zero loadings, simplifying the factor itself.
- **Quartimax**: Simplifies the rows of the factor loading matrix, making each variable load highly on only one factor.
- **Equamax**: A compromise between Varimax and Quartimax.

2. **Oblique Rotation** (Correlated Factors): The factor axes are not constrained to 90 degrees and are allowed to be correlated. This reflects the reality that most psychological and social constructs are related to some degree. This is generally the recommended default in EFA unless there's a strong theoretical reason for independent factors. If the oblique rotation shows very low correlations between factors, the results are essentially the same as an orthogonal rotation. Common methods are:

- **Direct Oblimin**: A common method where a parameter (Delta) controls the extent of the correlation.
- **Promax**: A method that first performs a Varimax rotation and then raises the loadings to a power to further simplify the structure, allowing the factors to become correlated.

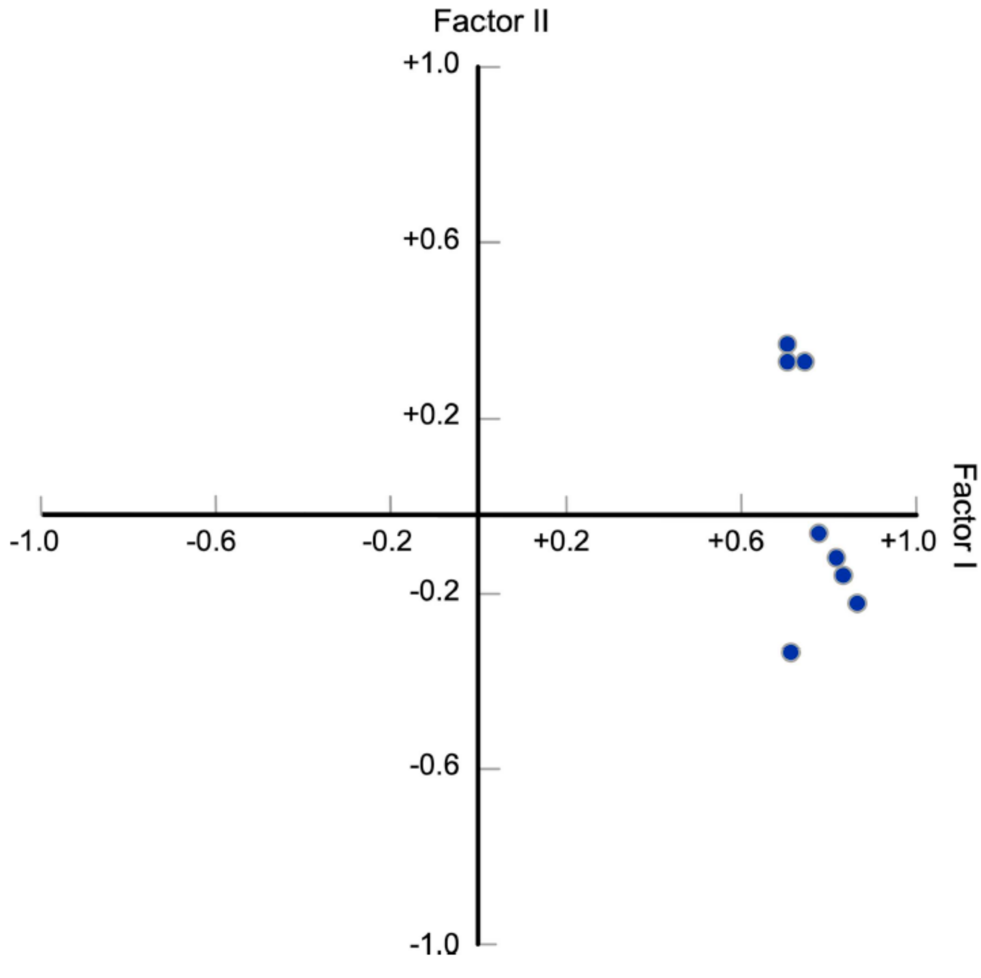



Figure 2: Unrotated Factor Solution

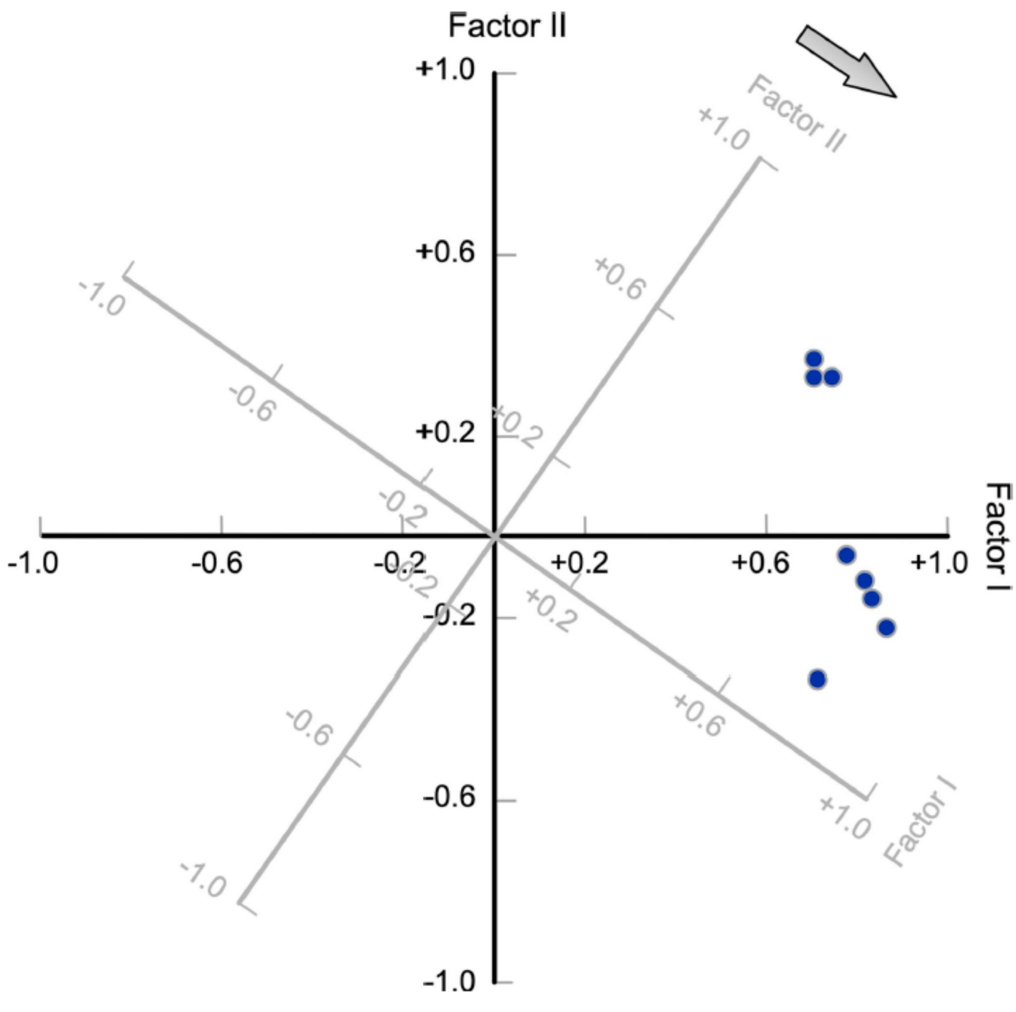



Figure 3: Orthogonally Rotated Factor Solution

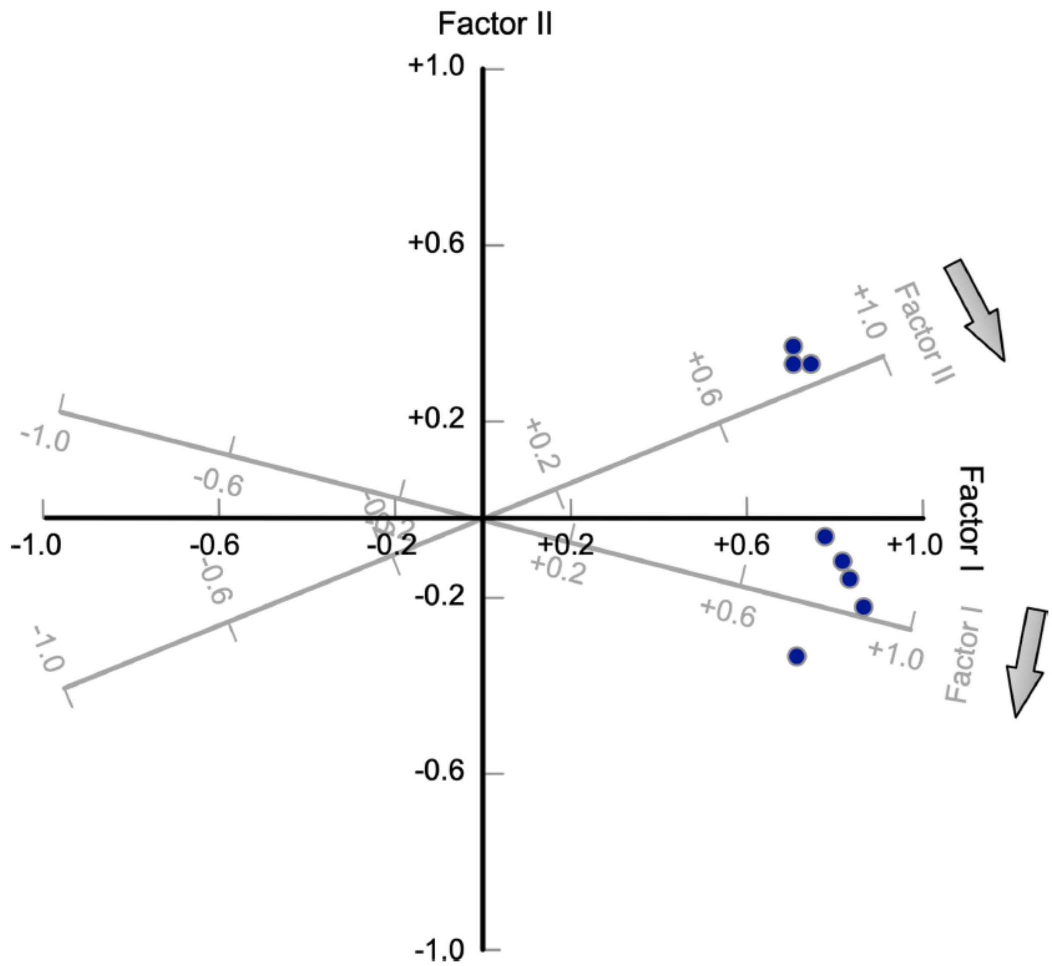



Figure 4: Obliquely Rotated Factor Solution

## Decision Steps in Exploratory Analysis

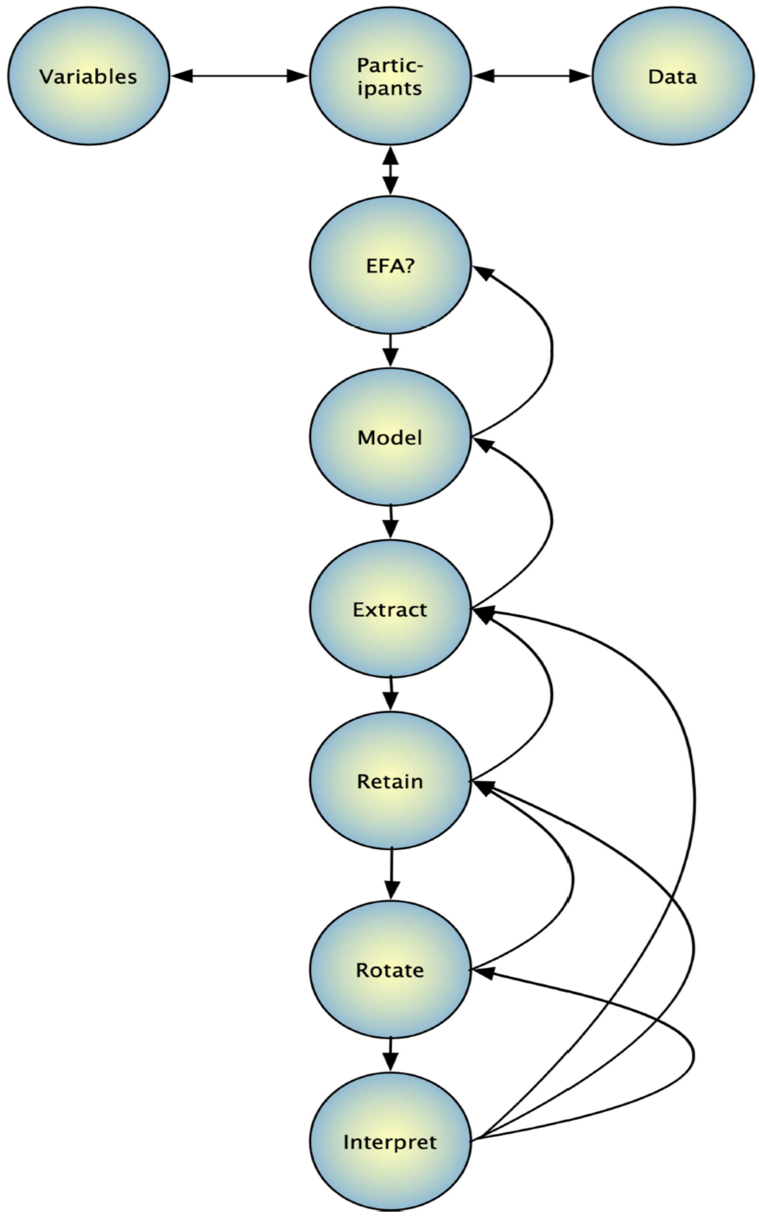



Figure 5: Flowchart of Decision Steps in Exploratory Factor Analysis

## Decision Steps in Exploratory Factor Analysis

**What variables to include**
- ☐ Number of variables per factor
- ☐ Adequate representation of the domain
- ☐ Avoid low communality
- ☐ Avoid low reliability
- ☐ Variables cannot be dependent upon each other

**What participants to include**
- ☐ Number of participants
- ☐ Adequate representation of the population

**Is data appropriate**
- ☐ Accuracy (out of range values, plausible summary values)
- ☐ Missing data (amount and distribution)
- ☐ Univariate and multivariate outliers
- ☐ Linearity
- ☐ Univariate and multivariate normality

**Is EFA appropriate**
- ☐ Bartlett's test of sphericity
- ☐ Kaiser-Meyer-Olkin test of sampling adequacy
- ☐ Correlation matrix

**Model of factor analysis**
- ☐ Principal components analysis
- ☐ Common factor analysis

**Factor extraction method**
- ☐ Weak factors/Nonnormal: Least-squares, Principal Axis
- ☐ Multivariate normal: Maximum Likelihood

**How many factors to retain**
- ☐ Parallel analysis
- ☐ Minimum average partials (MAP)
- ☐ Visual scree
- ☐ Theoretical convergence and parsimony
- ☐ Others

**Rotate factors**
- ☐ Orthogonal: Varimax
- ☐ Oblique: Oblimin, Promax

**Interpret results**
- ☐ Simple structure
- ☐ Theoretical convergence and parsimony

**Report results**
- ☐ All decision steps

Figure 6: Checklist of Decision Steps in Exploratory Factor Analysis

## Is Exploratory Factor Analysis Appropriate

Given that exploratory factor analysis (EFA) is based on the correlation matrix, it seems reasonable that the correlation matrix should contain enough covariance to justify conducting an EFA (Dziuban & Shirkey, 1974). First, the correlation matrix can be visually scanned to ensure that there are several coefficients $\geq 0.30$ (Hair et al., 2019; Tabachnick & Fidell, 2019). Scanning a large matrix for coefficients $\geq 0.30$ can be laborious, but can be visualized with a graph from the *qgraph* package.

A potential statistical problem can arise in EFA when correlations among the measured variables are too high. Singularity is the term used when measured variables are linearly dependent on each other like the verbal, nonverbal, and total (verbal + nonverbal) scores from a cognitive test. Extreme multicollinearity may also cause a problem during matrix computations akin to dividing by zero in arithmetic. Tabachnick and Fidell (2019) suggested that correlations above 0.90 among measured variables indicate multicollinearity. This may result in an error message (nonpositive definite matrix) from the software (Wothke, 1993).

Statistically, singularity prevents the matrix operations needed for EFA to be performed and multicollinearity causes those matrix operations to produce unstable results. The likelihood of a multicollinearity problem can be checked by ascertaining the determinant of the correlation matrix. If the determinant is greater than 0.00001 then multicollinearity is probably not a problem (Field et al., 2012).

A second assessment of the appropriateness of the correlation matrix for EFA is provided by Bartlett's test of sphericity (1950), which tests the null hypothesis that the correlation matrix is an identify matrix (ones on the diagonal and zeros on the off-diagonal) in the population. That is, it is random. Bartlett's test is sensitive to sample size and should be considered a minimal standard (Nunnally & Bernstein, 1994).

A final assessment of the appropriateness of the correlation matrix for EFA is offered by the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (Kaiser, 1974). KMO values range from 0 to 1, only reaching 1 when each variable is perfectly predicted by the other variables. A KMO value is a ratio of the sum of squared correlations to the sum of squared correlations plus the sum of squared partial correlations. Essentially, partial correlations will be small and KMO values large if the items share common variance. Kaiser (1974) suggested that KMO values $< 0.50$ are unacceptable but other measurement specialists recommended a minimum value of 0.60 (Mvududu & Sink, 2013; Watson, 2017) for acceptability with values $\geq 0.70$ preferred (Hoelzle & Meyer, 2013). KMO values for each variable as well as the total sample should be reviewed.

Bartlett's test and the KMO index can be computed with procedures included in the psych package after importing the iq data into RStudio.

### An example data set

The dataset labeled **iq**, contains scores from eight tests developed to measure cognitive ability that was administered to 152 participants. This dataset is in spreadsheet format and the file is labeled "iq.xls". These scores are from continuous variables with a mean of 100 and standard deviation of 15 in the general population. It was anticipated that four tests would measure verbal ability and four tests would measure nonverbal ability. The eight measured variables are described below.

| *Variable name* | *Description* |
|---|---|
| vocab1 | Provide correct definition of words |
| designs1 | Recreate square designs with colored blocks |
| similar1 | Describe how two words are similar |
| matrix1 | Complete matrix designs |
| veranal2 | Explain verbal analogies |
| vocab2 | Recognize correct definition of words |
| matrix2 | Recognize correct completion of matrix designs |
| designs2 | Recreate geometric designs with puzzle pieces |

```r
library(readxl)
library(qgraph)

iq <- read_excel("iq.xls")
head(iq)
```

```
## # A tibble: 6 x 8
##   vocab1 designs1 similar1 matrix1 veranal2 vocab2 matrix2 designs2
##    <dbl>    <dbl>    <dbl>   <dbl>    <dbl>  <dbl>   <dbl>    <dbl>
## 1    113      124      118     122      116    115     127      105
## 2    100       95      115      97      121    102     106      112
## 3    124      115      122     116      117    109     119      118
## 4    125      110      124     130      114    115     126      127
## 5    103       89      109      91      106    102     101       97
## 6     88      113       89     110       96     88     109      111
```

```r
cor(iq)
```

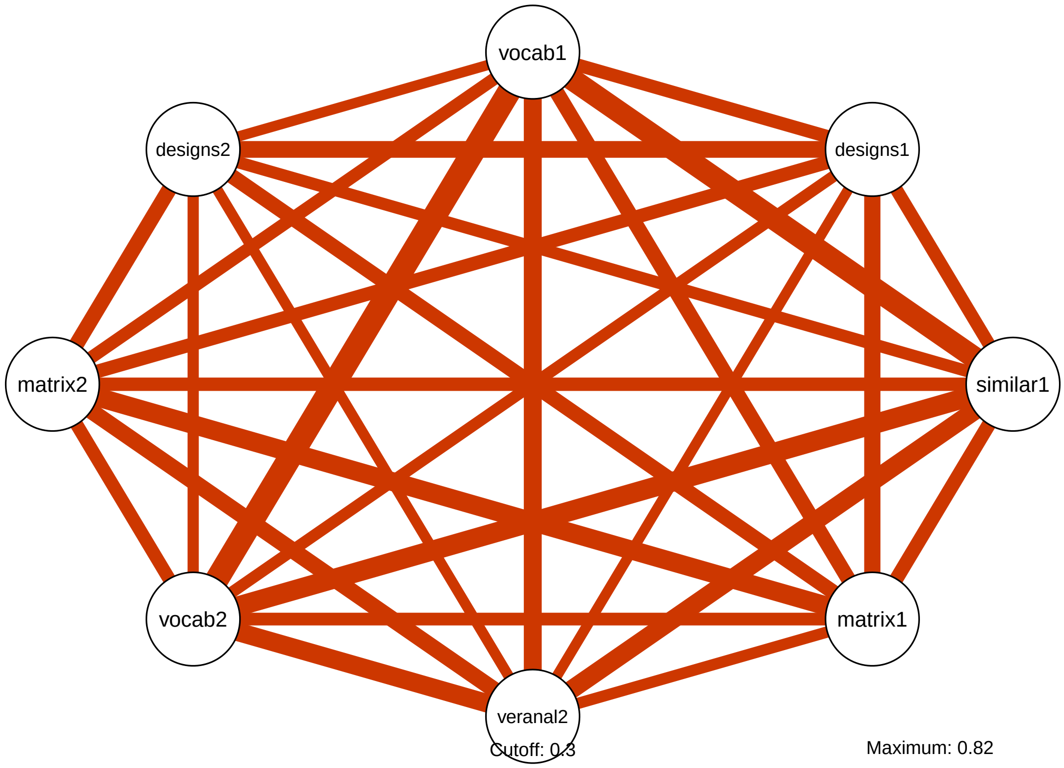
```
##             vocab1  designs1  similar1   matrix1  veranal2    vocab2   matrix2
## vocab1   1.0000000 0.5761746 0.7885174 0.6178617 0.6891966 0.8241553 0.5637402
## designs1 0.5761746 1.0000000 0.5697090 0.6499500 0.5086292 0.5389671 0.5862438
## similar1 0.7885174 0.5697090 1.0000000 0.5984887 0.7015425 0.7352893 0.5817558
## matrix1  0.6178617 0.6499500 0.5984887 1.0000000 0.5328058 0.5659428 0.7146091
```

```
## veranal2 0.6891966 0.5086292 0.7015425 0.5328058 1.0000000 0.7059415 0.6480372
## vocab2   0.8241553 0.5389671 0.7352893 0.5659428 0.7059415 1.0000000 0.5773067
## matrix2  0.5637402 0.5862438 0.5817558 0.7146091 0.6480372 0.5773067 1.0000000
## designs2 0.5073953 0.6615614 0.5504737 0.6189612 0.5137270 0.5302894 0.6244216
##           designs2
## vocab1   0.5073953
## designs1 0.6615614
## similar1 0.5504737
## matrix1  0.6189612
## veranal2 0.5137270
## vocab2   0.5302894
## matrix2  0.6244216
## designs2 1.0000000
```

```r
cor(iq) < 0.3
```

```
##          vocab1 designs1 similar1 matrix1 veranal2 vocab2 matrix2 designs2
## vocab1    FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## designs1  FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## similar1  FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## matrix1   FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## veranal2  FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## vocab2    FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## matrix2   FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
## designs2  FALSE    FALSE    FALSE   FALSE    FALSE  FALSE   FALSE    FALSE
```

```r
qgraph(cor(iq),
       cut=0.3,
       details = TRUE,
       posCol = "orangered3",
       negCol = "green",
       labels=names(iq))
```




```r
det(cor(iq))
```

```
## [1] 0.002444148
```

```r
library(psych)
library(FactoMineR)
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
##
## Attaching package: 'ggplot2'
```

```
## The following objects are masked from 'package:psych':
##
##     %+%, alpha
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WB
```

```r
KMO(iq)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = iq)
## Overall MSA =  0.9
## MSA for each item =
##   vocab1 designs1 similar1  matrix1 veranal2   vocab2  matrix2 designs2
##     0.87     0.92     0.93     0.90     0.92     0.90     0.89     0.91
```

```r
cortest.bartlett(iq)
```

```
## R was not square, finding R from data
```

```
## $chisq
## [1] 887.0736
##
## $p.value
## [1] 1.007166e-168
##
## $df
## [1] 28
```

## Factor Analysis Model

Two major models must be considered: **principal components analysis (PCA)** and **common factor analysis (EFA)**. Researchers sometimes claim that an EFA was conducted when a PCA model was actually applied (Osborne & Banjanovic,2016). However, PCA and EFA have different purposes and might, given the number and type of measured variables, produce different results. The purpose of EFA is to explain as well as possible the correlations among measured variables. In EFA, measured variables are thought to correlate with each other due to underlying latent constructs called factors. It also assumes that unique factors explain some variance beyond that explained by common factors. Conceptually, unique factors are composed of specific variance (systematic variance specific to a single measured variable) and error variance (unreliable error of measurement).

The purpose of PCA is to take the scores on a large set of observed variables and reduce them to scores on a smaller set of composite variables that retain as much information as possible from the original measured variables.

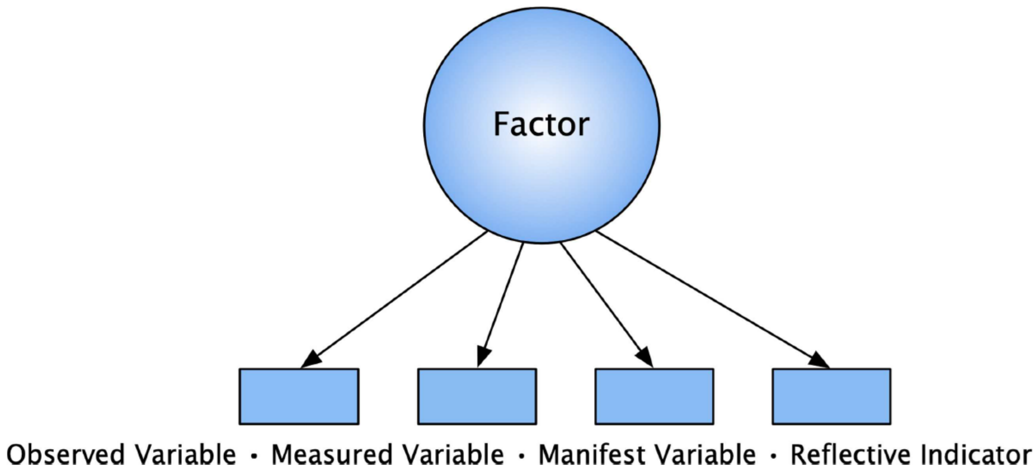



Figure 7: Common Factor Analysis Model

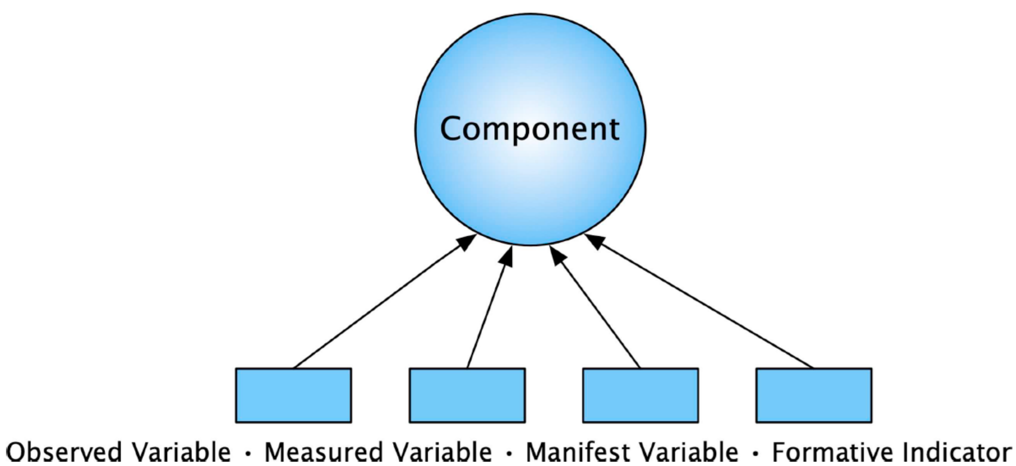



Figure 8: Principal Component Analysis Model

## Factor Extraction

After selecting the common factor model, the next step in EFA is to choose an extraction method. This might also be called the model fitting procedure or the parameter estimation procedure (Fabrigar & Wegener, 2012). In simple terms, this is the mathematical process of deriving the underlying factors from the correlation matrix.

Methodologists have devised a great number of extraction methods since Spearman (1904) first imagined EFA. These include image analysis, alpha analysis, non-iterated principal axis (PA), iterated principal axis (IPA), maximum likelihood (ML), unweighted least squares (ULS), weighted least squares (WLS), generalized least squares (GLS), minimum residual (MINRES), etc. Some extraction methods are identical yet were given different names. For example, ULS, OLS, and MINRES all refer to the same extraction method (Flora, 2018). Further, an IPA extraction will converge to an OLS solution (Briggs & MacCallum, 2003; MacCallum, 2009). Mathematically, extraction methods differ in the way they go about locating the factors that will best reproduce the original correlation matrix, whether they attempt to reproduce the sample correlation matrix or the population correlation matrix, and in their definition of the best way to measure closeness of the reproduced and original correlation matrices.

Simulation research has compared ML and least squares extraction methods in terms of factor recovery under varying levels of sample size and factor strength (Briggs & MacCallum, 2003; de Winter & Dodou, 2012; MacCallum et al., 2007; Ximenez, 2009). ML may be appropriate for larger samples with normal data and strong factors, whereas least squares methods may be preferable for smaller samples with nonnormal data or weak factors (MacCallum et al., 2007; Watson, 2017).

Regardless of extraction method, researchers must be cautious of improper solutions. That is, solutions that are mathematically impossible. For example, communality values greater than 1.00. Researchers must also be aware that extraction may fail with iterative estimators such as IPA. This happens because EFA software will try to arrive at an optimal estimate of the correlation matrix before some maximum number of estimation iterations has been completed. If an optimal estimate has not been computed at that point, an error message mentioning nonconvergence will be produced. In that case, the researcher may increase the maximum number of iterations and rerun the analysis. If nonconvergence persists after >100 iterations then the results, like those from an improper solution, should not be interpreted (Flora, 2018). "Improper solutions and nonconvergence are more likely to occur when there is a linear dependence among observed variables, when the model includes too many common factors, or when the sample size is too small" (Flora, 2018, p. 257). Alternatively, another extraction method could be employed (i.e., a sensitivity analysis) to ensure that the results are robust to extraction method.

## How Many Factors to Retain

As many factors as measured variables can be extracted for exploratory factor analysis (EFA), but it is usually possible to explain the majority of variance with a smaller number of factors.

The problem arises in determining the exact number of factors to retain for interpretation. Methodologists have observed that this is probably the most important decision in EFA because there are serious consequences for selecting either too few or too many factors (Benson & Nasser, 1998; Fabrigar et al., 1999; Glorfeld, 1995; Hoelzle & Meyer, 2013; Preacher et al., 2013), whereas the options for other EFA decisions tend to be fairly robust (Hayton et al., 2004; Tabachnick & Fidell, 2019). Retaining too few factors can distort factor loadings and result in solutions in which common factors are combined, thereby obscuring the true factor solution. Extracting too many factors can focus on small, unimportant factors that are difficult to interpret and unlikely to replicate (Hayton et al., 2004). "Choosing the number of factors is something like focusing a microscope. Too high or too low an adjustment will obscure a structure that is obvious when the adjustment is just right" (Hair et al., 2019, p. 144).

Current best practices advocate using a combination of the following methods:

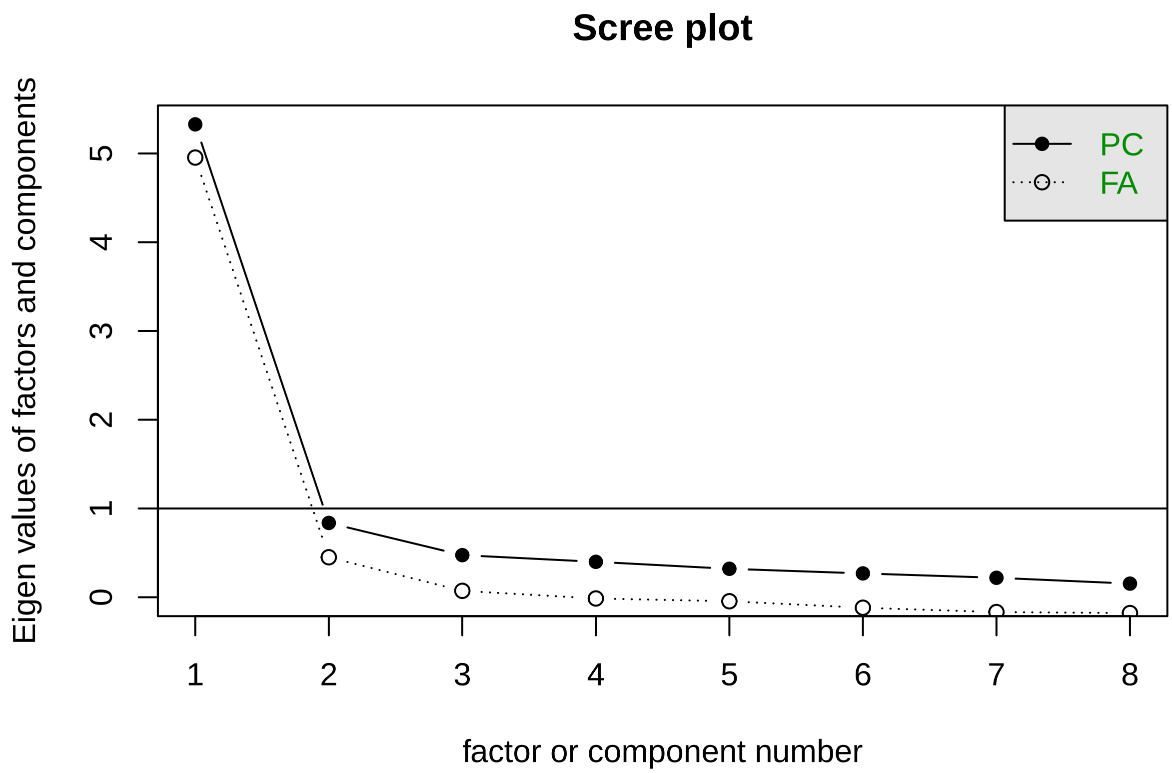
- Parallel Analysis (PA): This is widely considered the best available and most accurate method. It compares the actual eigenvalues from your data to eigenvalues generated from random data of the same sample size and number of variables. You retain only factors whose observed eigenvalues are larger than the corresponding random data eigenvalues.
- Minimum Average Partial (MAP) Test: The MAP test, developed by Velicer (1976), is also highly accurate. It involves calculating the average squared partial correlation among variables after sequentially partialing out factors. The number of factors that results in the minimum average partial correlation is the optimal number to retain.
- Cattell's Scree Test: This is a visual method using a scree plot, which graphs each eigenvalue. You visually examine the plot for an "elbow" or the point where the line begins to flatten significantly. Factors above this elbow are retained. While subjective, it is a useful supplement to objective methods.
- Kaiser Criterion (Eigenvalue > 1.0): This method is the most popular in practice, but experts widely agree it is among the least accurate and tends to overestimate the number of factors. The rule was originally proposed for Principal Component Analysis (PCA) and is problematic for EFA. It is best used as a supportive indicator and not the sole determinant.

Researchers are encouraged to use multiple criteria and consider the theoretical meaningfulness of the factors in their specific research context. A consensus across several methods (e.g., PA suggesting 3 factors, MAP suggesting 3, and the Scree plot showing an elbow at 3) provides strong empirical justification.

```r
(eigen.iq <- eigen(cor(iq)))
```

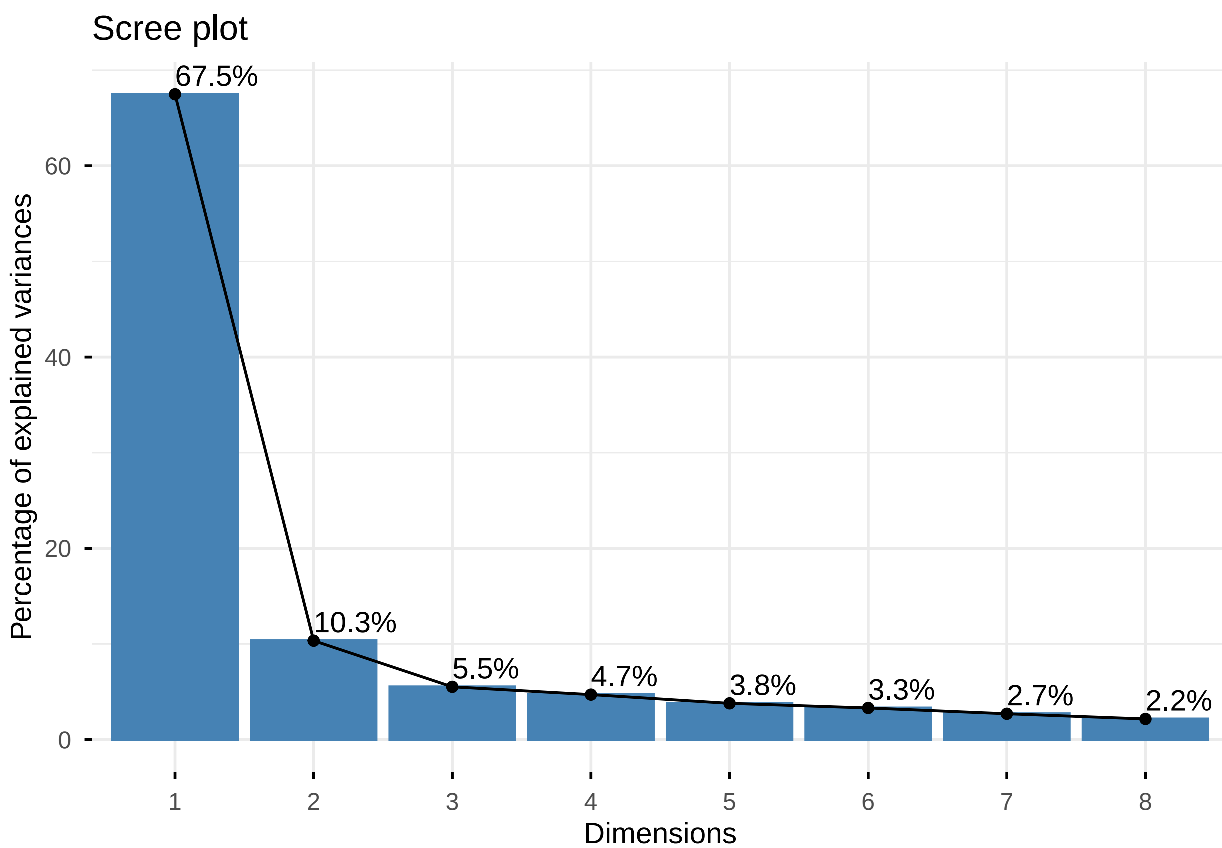
```
## eigen() decomposition
## $values
## [1] 5.3280352 0.8370123 0.4743501 0.3989081 0.3203761 0.2683994 0.2193804
## [8] 0.1535386
##
## $vectors
##            [,1]       [,2]       [,3]         [,4]        [,5]        [,6]
## [1,] -0.3721036 -0.3721233  0.2241728  0.245813913  0.11208776 -0.14053127
## [2,] -0.3356323  0.3875447  0.5107787  0.042060533 -0.66983400 -0.09777802
## [3,] -0.3688554 -0.2988752  0.1523806  0.020370333  0.09630923  0.75578266
## [4,] -0.3508000  0.3339271 -0.1688658  0.637677449  0.19399642  0.11187011
## [5,] -0.3529295 -0.2917982 -0.3974312 -0.429376564 -0.42276693  0.08529879
## [6,] -0.3658676 -0.3822132  0.1270448 -0.005269801  0.20761926 -0.59950067
## [7,] -0.3505881  0.2704114 -0.6474433  0.029210696 -0.05004396 -0.14132651
## [8,] -0.3293092  0.4540309  0.2093834 -0.587804120  0.51721044  0.01783103
##             [,7]        [,8]
## [1,]  0.02388944  0.76125014
## [2,]  0.11167351 -0.06153801
## [3,]  0.31873173 -0.26251040
## [4,] -0.50562057 -0.15646780
## [5,] -0.51066354  0.03455150
## [6,]  0.06793561 -0.54476027
## [7,]  0.59201421  0.10474309
## [8,] -0.11765597  0.11995312
```

```r
scree(iq, pc=TRUE, fa=TRUE)
```




```r
fviz_eig(prcomp(iq),addlabels = TRUE)
```

```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
```

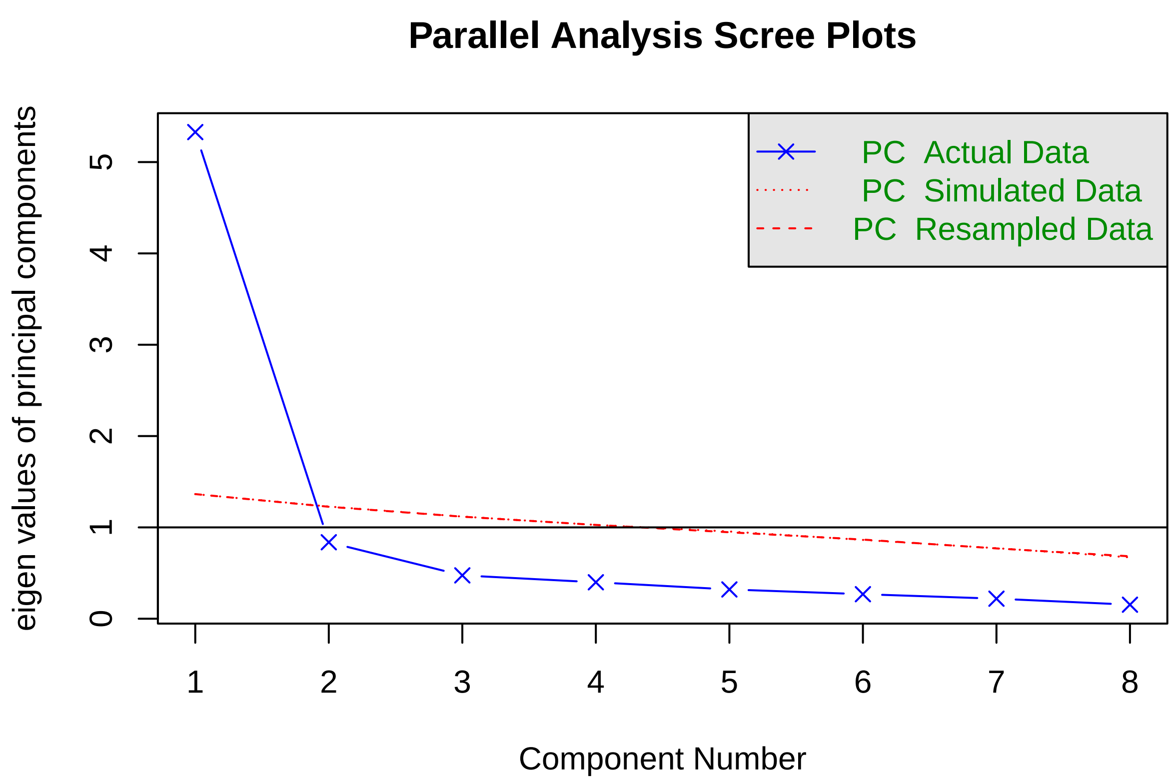



```r
(map.iq <- VSS(iq, plot=FALSE))
```

```
##
## Very Simple Structure
## Call: vss(x = x, n = n, rotate = rotate, diagonal = diagonal, fm = fm,
##     n.obs = n.obs, plot = plot, title = title, use = use, cor = cor)
## VSS complexity 1 achieves a maximimum of 0.95  with  1  factors
## VSS complexity 2 achieves a maximimum of 0.97  with  2  factors
##
## The Velicer MAP achieves a minimum of 0.06  with  2  factors
## BIC achieves a minimum of  -33.6  with  2  factors
## Sample Size adjusted BIC achieves a minimum of  -2.51  with  4  factors
##
## Statistics by number of factors
##   vss1 vss2   map dof   chisq     prob sqresid  fit RMSEA   BIC SABIC complex
## 1 0.95 0.00 0.070  20 1.2e+02 2.1e-16    1.48 0.95 0.182  20.2  83.5     1.0
## 2 0.59 0.97 0.063  13 3.2e+01 2.7e-03    0.83 0.97 0.097 -33.6   7.5     1.5
## 3 0.50 0.84 0.095   7 1.1e+01 1.3e-01    0.62 0.98 0.063 -23.9  -1.7     1.8
## 4 0.49 0.79 0.170   2 1.2e+00 5.5e-01    0.53 0.98 0.000  -8.8  -2.5     1.9
## 5 0.37 0.69 0.324  -2 3.0e-07       NA    0.38 0.99    NA    NA    NA     2.2
## 6 0.37 0.75 0.510  -5 8.9e-08       NA    0.39 0.99    NA    NA    NA     2.2
```

```
## 7 0.37 0.75 1.000  -7 0.0e+00      NA     0.40 0.99    NA    NA    NA      2.2
## 8 0.37 0.75    NA  -8 0.0e+00      NA     0.40 0.99    NA    NA    NA      2.2
##    eChisq    SRMR eCRMS  eBIC
## 1 4.3e+01 7.1e-02 0.084 -57.3
## 2 6.4e+00 2.7e-02 0.040 -58.9
## 3 1.6e+00 1.4e-02 0.027 -33.6
## 4 1.6e-01 4.3e-03 0.016  -9.9
## 5 4.8e-08 2.4e-06    NA    NA
## 6 8.2e-09 9.8e-07    NA    NA
## 7 8.0e-18 3.1e-11    NA    NA
## 8 8.0e-18 3.1e-11    NA    NA
```

```
(para.iq <- fa.parallel(iq, fa="pc"))
```




```
## Parallel analysis suggests that the number of factors =  NA  and the number of compon

## Call: fa.parallel(x = iq, fa = "pc")
## Parallel analysis suggests that the number of factors =  NA  and the number of compon
##
##  Eigen Values of
```

```
##
##  eigen values of factors
## [1]  4.95  0.45  0.07 -0.01 -0.05 -0.12 -0.17 -0.18
##
##  eigen values of simulated factors
## [1] NA
##
##  eigen values of components
## [1] 5.33 0.84 0.47 0.40 0.32 0.27 0.22 0.15
##
##  eigen values of simulated components
## [1] 1.36 1.23 1.12 1.03 0.95 0.86 0.77 0.67
```

```r
fa2.result <- fa(iq, nfactors = 2, fm = "ml", rotate = "promax")
```

```
## Loading required namespace: GPArotation
```

```r
print(fa2.result, digits = 3, sort = TRUE)
```

```
## Factor Analysis using method =  ml
## Call: fa(r = iq, nfactors = 2, rotate = "promax", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matrix
##          item    ML1    ML2    h2    u2  com
## vocab1      1  0.959 -0.047 0.855 0.145 1.00
## vocab2      6  0.896 -0.011 0.787 0.213 1.00
## similar1    3  0.752  0.130 0.731 0.269 1.06
## veranal2    5  0.616  0.211 0.621 0.379 1.23
## designs2    8 -0.024  0.799 0.611 0.389 1.00
## matrix2     7  0.056  0.779 0.677 0.323 1.01
## matrix1     4  0.077  0.770 0.688 0.312 1.02
## designs1    2  0.088  0.705 0.598 0.402 1.03
##
##                         ML1   ML2
## SS loadings           2.926 2.641
## Proportion Var        0.366 0.330
## Cumulative Var        0.366 0.696
## Proportion Explained  0.526 0.474
## Cumulative Proportion 0.526 1.000
##
##  With factor correlations of
##       ML1   ML2
## ML1 1.000 0.758
## ML2 0.758 1.000
```

```
##
## Mean item complexity =  1
## Test of the hypothesis that 2 factors are sufficient.
##
## df null model =  28  with the objective function =  6.014 with Chi Square =  887.074
## df of  the model are 13  and the objective function was  0.212
##
## The root mean square of the residuals (RMSR) is  0.028
## The df corrected root mean square of the residuals is  0.041
##
## The harmonic n.obs is  152 with the empirical chi square  6.8  with prob <  0.912
## The total n.obs was  152  with Likelihood Chi Square =  30.994  with prob <  0.00338
##
## Tucker Lewis Index of factoring reliability =  0.9545
## RMSEA index =  0.0952  and the 90 % confidence intervals are  0.0525 0.1396
## BIC =  -34.316
## Fit based upon off diagonal values = 0.998
## Measures of factor score adequacy
##                                                    ML1   ML2
## Correlation of (regression) scores with factors   0.967 0.944
## Multiple R square of scores with factors          0.935 0.891
## Minimum correlation of possible factor scores     0.871 0.781
```

```
#Compute Cronbach alpha
library(tidyverse)
f1 <- iq %>%
  select(vocab1, vocab2, similar1, veranal2) %>%
  as.data.frame()

f2 <- iq %>%
  select(designs2, matrix2, matrix1, designs1) %>%
  as.data.frame()

psych::alpha(iq)
```

```
##
## Reliability analysis
## Call: psych::alpha(x = iq)
##
##   raw_alpha std.alpha G6(smc) average_r S/N    ase mean sd median_r
##       0.93      0.93    0.93      0.62  13 0.0088  100 13     0.59
##
##     95% confidence boundaries
##          lower alpha upper
```

```
## Feldt     0.91  0.93  0.94
## Duhachek  0.91  0.93  0.95
##
##  Reliability if an item is dropped:
##          raw_alpha std.alpha G6(smc) average_r S/N alpha se  var.r med.r
## vocab1        0.91      0.91    0.92      0.61  11   0.0106 0.0048  0.59
## designs1      0.92      0.92    0.93      0.63  12   0.0096 0.0082  0.62
## similar1      0.92      0.92    0.92      0.61  11   0.0105 0.0066  0.59
## matrix1       0.92      0.92    0.92      0.62  11   0.0099 0.0086  0.58
## veranal2      0.92      0.92    0.92      0.62  11   0.0101 0.0071  0.59
## vocab2        0.92      0.92    0.92      0.61  11   0.0104 0.0055  0.60
## matrix2       0.92      0.92    0.92      0.62  11   0.0099 0.0088  0.60
## designs2      0.92      0.92    0.93      0.63  12   0.0095 0.0076  0.60
##
##  Item statistics
##            n raw.r std.r r.cor r.drop mean sd
## vocab1   152  0.86  0.85  0.85   0.81   98 17
## designs1 152  0.77  0.78  0.74   0.71   98 14
## similar1 152  0.85  0.85  0.83   0.80  104 17
## matrix1  152  0.81  0.81  0.78   0.75  100 17
## veranal2 152  0.81  0.81  0.78   0.75  102 15
## vocab2   152  0.85  0.84  0.83   0.79  101 16
## matrix2  152  0.81  0.81  0.78   0.74  101 16
## designs2 152  0.76  0.77  0.72   0.69  101 14
```

```r
psych::alpha(f1)
```

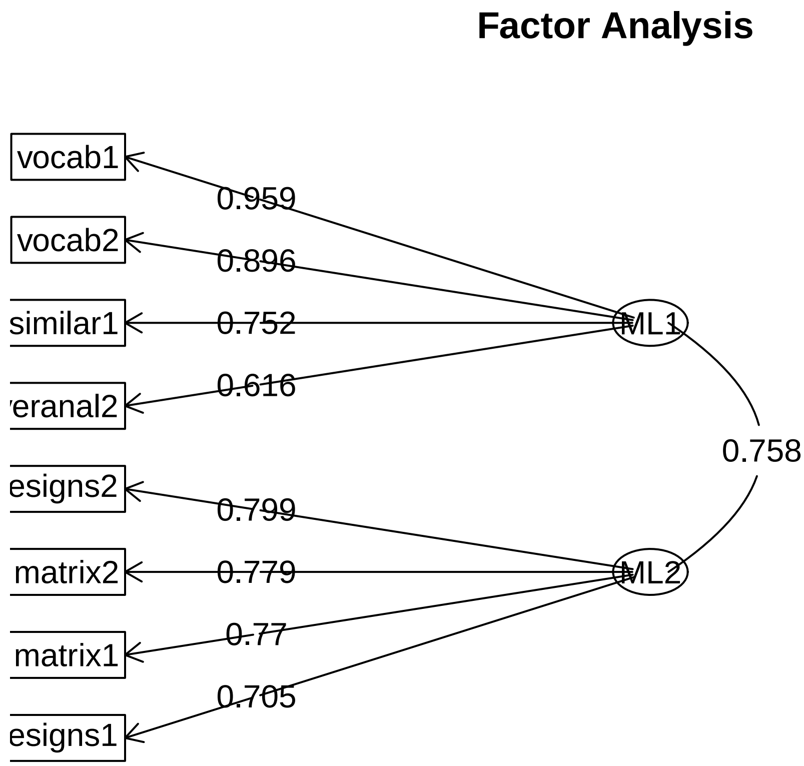
```
##
## Reliability analysis
## Call: psych::alpha(x = f1)
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean sd median_r
##       0.92      0.92     0.9      0.74  11 0.011  101 15     0.72
##
##     95% confidence boundaries
##          lower alpha upper
## Feldt      0.9  0.92  0.94
## Duhachek   0.9  0.92  0.94
##
##  Reliability if an item is dropped:
##          raw_alpha std.alpha G6(smc) average_r  S/N alpha se   var.r med.r
## vocab1        0.88      0.88    0.83      0.71  7.5    0.017 0.00034  0.71
## vocab2        0.89      0.89    0.85      0.73  8.0    0.015 0.00293  0.70
## similar1      0.89      0.90    0.86      0.74  8.5    0.015 0.00541  0.71
```

```
## veranal2      0.92      0.92    0.88      0.78 10.8    0.012 0.00200  0.79
##
##  Item statistics
##            n raw.r std.r r.cor r.drop mean sd
## vocab1   152  0.92  0.92  0.90   0.86   98 17
## vocab2   152  0.91  0.91  0.88   0.84  101 16
## similar1 152  0.90  0.90  0.85   0.82  104 17
## veranal2 152  0.85  0.86  0.78   0.76  102 15
```

```r
psych::alpha(f2)
```

```
##
## Reliability analysis
## Call: psych::alpha(x = f2)
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean sd median_r
##       0.88      0.88    0.85      0.64 7.2 0.016  100 13     0.64
##
##     95% confidence boundaries
##          lower alpha upper
## Feldt     0.84  0.88  0.91
## Duhachek  0.84  0.88  0.91
##
##  Reliability if an item is dropped:
##          raw_alpha std.alpha G6(smc) average_r S/N alpha se   var.r med.r
## designs2      0.85      0.85    0.79      0.65 5.6    0.021 0.00412  0.65
## matrix2       0.84      0.84    0.78      0.64 5.4    0.022 0.00048  0.65
## matrix1       0.83      0.83    0.77      0.62 5.0    0.024 0.00142  0.62
## designs1      0.85      0.85    0.79      0.65 5.6    0.021 0.00289  0.62
##
##  Item statistics
##            n raw.r std.r r.cor r.drop mean sd
## designs2 152  0.84  0.85  0.77   0.72  101 14
## matrix2  152  0.86  0.85  0.79   0.74  101 16
## matrix1  152  0.88  0.87  0.82   0.77  100 17
## designs1 152  0.84  0.85  0.77   0.72   98 14
```

```r
#Path diagram
fa.diagram(fa2.result, sort = TRUE, digits = 3, cex = 0.5, cut = 0)
```




## Factor Names

For convenience, researchers typically name the factors identified through EFA. Names can be given to factors to facilitate the communication of results and advance cumulative knowledge.

The four measured variables that saliently loaded on the first factor contained verbal content and seemed to require reasoning with that content: defining words, describing how words are similar, and explaining verbal analogies. The four measured variables that saliently loaded on the second factor contained nonverbal content that involved analysis for patterns, inductive discovery of relationships, and deductive identification of missing components. Thus, the first factor might be called verbal reasoning and the second nonverbal reasoning.